

DW01_absval

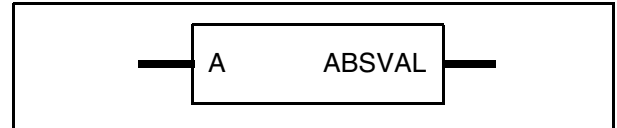
Absolute Value

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameterized word length
- Inferable through a function call

Revision History



Description

DW01_absval determines the absolute value of data from input port A and places this value on output port ABSVAL.

The input is assumed to be represented as a signed two's complement number. The output is represented as an unsigned number.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
A	<i>width</i> bits	Input	Input data
ABSVAL	<i>width</i> bits	Output	Absolute value of A

Table 1-2 Parameter Description

Parameter	Values	Function
width	≥ 1	Word length of A and ABSVAL

Table 1-3 Synthesis Implementations^a

Implementation Name	Function	License Feature Required
rpl	Ripple-carry synthesis model	none
cla	Carry-look-ahead synthesis model	none
clf	Fast carry-look-ahead synthesis model	DesignWare

- a. During synthesis, Synopsys synthesis tools select the appropriate architecture for your constraints. Alternatively, you may use the `set_implementation` command to choose one specific implementation from those listed in this table. For details, see the documentation for your Synopsys synthesis tool.

Table 1-4 Simulation Models

Model	Function
DW01.DW01_ABSVAL_CFG_SIM	Design unit name for VHDL simulation
dw/dw01/src/DW01_absval_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW01_absval.v	Verilog simulation model source code

Table 1-5 Truth Table (*width* = 3)

A(2:0)	Value of A	ABSVAL(2:0)	Value of ABSVAL
000	0	000	0
001	1	001	1
010	2	010	2
011	3	011	3
100	−4	100	4 ^a
101	−3	011	3
110	−2	010	2
111	−1	001	1

a. Only correct if the result is interpreted as an unsigned number.

Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Function Inferencing - VHDL

```
library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_arith.all;
use DWARE.DW_foundation_arith.all;

entity DW01_absval_func is
    generic(width: integer:= 8);
    port( func_A : in std_logic_vector(width-1 downto 0);
          ABSVAL_func : out std_logic_vector(width-1 downto 0) );
end DW01_absval_func;

architecture func of DW01_absval_func is
begin

    ABSVAL_func <= std_logic_vector(DWF_absval(SIGNED(func_A)));
end func;
```

HDL Usage Through Function Inferencing - Verilog

```
module DW01_absval_func (func_A, ABSVAL_func);
    parameter func_A_width = 8;

    // Passes the width to the absolute value function
    parameter width = func_A_width;

    // Please add search_path = search_path + {synopsys_root + "/dw/sim_ver"}
    // to your .synopsys_dc.setup file (for synthesis) and add
    // +incdir+$SYNOPSYS/dw/sim_ver+ to your verilog simulator command line
    // (for simulation).
    `include "DW01_absval_function.inc"

    input  [func_A_width-1 : 0] func_A;
    output [func_A_width-1 : 0] ABSVAL_func;
    assign ABSVAL_func = DWF_absval(func_A);

endmodule
```

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW01_absval_inst is
  generic ( inst_width : NATURAL := 8 );
  port ( inst_A : in std_logic_vector(inst_width-1 downto 0);
        ABSVAL_inst : out std_logic_vector(inst_width-1 downto 0) );
end DW01_absval_inst;

architecture inst of DW01_absval_inst is
begin

  -- Instance of DW01_absval
  U1 : DW01_absval
    generic map ( width => inst_width )
    port map ( A => inst_A, ABSVAL => ABSVAL_inst );
end inst;

-- pragma translate_off
configuration DW01_absval_inst_cfg_inst of DW01_absval_inst is
  for inst
  end for; -- inst
end DW01_absval_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW01_absval_inst( inst_A, ABSVAL_inst );
  parameter width = 8;

  input [width-1 : 0] inst_A;
  output [width-1 : 0] ABSVAL_inst;

  // Instance of DW01_absval
  DW01_absval #(width)
    U1 ( .A(inst_A), .ABSVAL(ABSVAL_inst) );
endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 4 ■ Added this Revision History table and the document links on this page

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